Automated Personal Protective Equipment Detection Using Deep Learning Object Detection

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ABSTRACT

This paper presents an AI-assisted Personal Protective Equipment (PPE) detection system for safety monitoring in construction and industrial environments. The proposed solution uses a YOLOv8n-based object detection pipeline to identify both compliance and non-compliance classes, including Hardhat, Safety Vest, Mask, NO-Hardhat, and NO-Mask, from images, video streams, and real-time webcam input. To improve operational usability, the system integrates a natural language module that generates concise detection summaries and safety alerts from model outputs. In addition, a reinforcement learning component is included as a policy-learning prototype for alert decision support. Experimental evaluation was conducted using baseline and ablation configurations on a public PPE dataset. Results indicate that increasing the training data fraction improves mean Average Precision (mAP), while higher input resolution yields higher precision but limited recall gains. Across runs, recall remains the primary performance bottleneck, suggesting that the current model may under-detect challenging or minority classes under practical conditions. The project demonstrates an end-to-end, reproducible implementation that satisfies requirements for deep learning, NLP integration, RL experimentation, and ethical documentation. While current performance supports the practical feasibility of assisted monitoring, the system is most suitable as a decision-support tool with human-in-the-loop review, rather than as a fully autonomous enforcement mechanism.

1. INTRODUCTION

1.1 Problem Statement

Workplace safety is a critical concern in construction sites, factories, and industrial environments. Workers are required to wear Personal Protective Equipment (PPE) such as hard hats, safety vests, and face masks to reduce the risk of injuries and accidents. However, monitoring PPE compliance is often done manually by supervisors, which can be inconsistent, time-consuming, and prone to human error. As a result, safety violations may go unnoticed, increasing the likelihood of workplace incidents.

Recent advances in computer vision and deep learning have enabled automated systems capable of detecting objects in images and videos. Object detection models such as YOLO (You Only Look Once) have demonstrated strong performance in real-time detection tasks. These models use convolutional neural networks (CNNs) to identify objects within an image and draw bounding boxes around them. Applying these technologies to workplace safety monitoring can help automate the detection of PPE usage and assist safety officers in identifying violations more efficiently.

This project proposes a PPE detection system that uses deep learning-based object detection models to identify safety equipment in worksite images. The system will detect three PPE categories: hardhats, safety vests, and face masks. The objective is to build a model that can automatically detect whether PPE equipment is present in images taken from workplace environments.

1.2 Motivation

Ensuring consistent PPE compliance in high-risk workplaces is both essential and difficult to sustain through manual supervision alone. In construction and industrial settings, safety officers must monitor many workers across dynamic and visually complex scenes, often under time pressure. This creates a gap between safety policy and day-to-day enforcement, where missed observations can lead to preventable injuries. The motivation for this work is to reduce that gap by providing an AI-assisted monitoring tool that improves the speed, consistency, and coverage of PPE checks.

Recent progress in real-time object detection makes automated PPE monitoring technically feasible, but practical deployment requires more than raw detection accuracy. Safety teams need outputs that are understandable, actionable, and aligned with operational workflows. For this reason, the project combines three complementary components: (1) a CNN-based YOLOv8n detector for recognizing PPE and non-compliance classes, (2) an NLP layer that translates detections into concise summaries and alerts, and (3) an RL-based prototype for exploring smarter alert decision

policies. Together, these components are motivated by a real-world requirement: supporting human safety officers with faster and clearer decision support rather than replacing them.

Another key motivation is to build a reproducible and ethically grounded pipeline that can be evaluated and improved over time. The ablation findings show that performance trade-offs remain, especially the persistent recall bottleneck, which is critical in safety applications because missed detections can have serious consequences. This reinforces the need for a human-in-the-loop design, where AI improves monitoring efficiency while final judgment remains with trained personnel. Therefore, the project is motivated not only by technical advancement, but also by the practical goal of delivering a responsible, assistive safety system suitable for real operational environments.

1.3 Scope

The scope of the project focuses on developing and evaluating an object detection model trained on a labeled PPE dataset. The model will be tested using images containing workers wearing or not wearing safety equipment.

1.3 Success Metrics

The performance of the system is evaluated using standard object detection metrics to ensure both accuracy and efficiency are properly assessed.

Mean Average Precision at IoU threshold 0.5 (mAP@0.5) is used as the primary metric for measuring detection accuracy. This metric evaluates how well the predicted bounding boxes overlap with the ground truth annotations using an Intersection over Union (IoU) threshold of 0.5. For each class, the Average Precision (AP) is computed by integrating the precision–recall curve, and the mean of these values across all classes yields the mAP. A higher mAP@0.5 indicates better localization and classification performance of the model.

Precision and Recall are used to assess the reliability of detections. Precision measures the proportion of correctly predicted positive detections out of all predicted positives, indicating how accurate the model's detections are. Recall, on the other hand, measures the proportion of correctly detected objects out of all actual objects present in the dataset, reflecting the model's ability to find all relevant instances. Together, these metrics provide insight into the trade-off between false positives and false negatives.

Precision–Recall (PR) curves are utilized to visualize the relationship between precision and recall across different confidence thresholds. These curves illustrate how the model's performance changes as the detection threshold varies, helping identify the optimal balance between precision and recall. A curve that remains closer to the top-right corner indicates stronger overall performance.

Latency is measured to evaluate the efficiency of the system during inference. It represents the time taken by the model to process an input and produce detection results. Low latency is crucial for real-time applications, as it ensures that detections are performed quickly without significant delay. This metric is typically measured in milliseconds per image or frames per second (FPS), providing insight into the system's practical deployment capability.

1.4 Constraints

The implementation of the proposed safety monitoring system is subject to several technical and environmental constraints:

Dataset generalization remains a key limitation, as the relatively limited size of available datasets may affect the model's ability to generalize across diverse workplace settings. **Hardware resources** also impose constraints, potentially limiting the total training duration and restricting the architectural complexity of the neural network. Additionally, **environmental variability**, including differences in ambient lighting, camera perspectives, and worker orientations, may influence the overall detection accuracy of the system.

Despite these identified limitations, the project is designed to deliver a functional PPE detection prototype that demonstrates the feasibility of identifying safety equipment within dynamic worksite imagery.

2. DATASET AND PREPROCESSING

The dataset used in this project will consist of labeled images containing workers wearing personal protective equipment (PPE) in workplace environments. Each image contains bounding box annotations identifying specific PPE items. The primary object categories considered in this study are hard hats, safety vests, and face masks, which are among the most commonly required safety equipment in construction and industrial workplaces. These annotations enable the training of an object detection model capable of recognizing PPE equipment in images.

2.1 Dataset Source

The dataset used in this project consists of labeled images capturing workplace environments with various safety conditions and equipment. Each image includes bounding box annotations that identify multiple object classes related to personal protective equipment (PPE) and safety monitoring [6]. The object categories considered in this study include Hardhat, Mask, Safety Vest, No Safety Vest, Goggles, Safety Cone, Ladder, and Fall Detected. These classes were selected to represent both compliance (presence of PPE) and violations or hazards (absence of PPE and unsafe conditions). The annotated data enables the training of an object detection model capable of recognizing PPE usage, detecting missing equipment, and identifying potential safety risks in real time.

2.2 License

The dataset used in this study is publicly available through Roboflow Universe, a platform that hosts datasets for research and educational purposes. Specifically, the Personal Protective Equipment Combined Model dataset is distributed under the Creative Commons Attribution 4.0 International License, which permits use, sharing, and adaptation of the data provided that appropriate credit is given to the original creators [6].

In compliance with this license, the project utilizes the dataset strictly for academic purposes and ensures that proper attribution is provided to the dataset source. All usage adheres to the licensing terms specified by the dataset provider.

2.3 Data Collection Notes

The dataset was sourced from the Roboflow PPE Combined v8 collection and used as a supervised object detection dataset for workplace safety scenarios. It contains annotated images from construction and industrial-like environments with varied camera angles, lighting conditions, and scene complexity. The label space

includes both compliance classes (for example, Hardhat, Mask, Safety Vest) and non-compliance classes (for example, NO-Hardhat, NO-Mask, NO-Safety Vest), which supports violation-aware monitoring. Because the dataset is public and externally curated, this project inherits its collection limitations, including possible class imbalance, domain bias, and uneven representation of difficult edge cases such as occlusion and small objects.

2.4 Data Preprocessing

Data was prepared in YOLO format with image-label pairing validation to ensure each training sample had consistent annotation files. The pipeline used standard YOLO preprocessing behavior, including resizing to a fixed input resolution and normalization handled by the training framework. During experiments, image size and data fraction were varied as part of ablation design to measure sensitivity to input scale and effective data volume. Quality checks were also performed to detect missing or orphan labels, reducing silent training failures from malformed dataset structure.

2.5 Data Splits

The project used fixed train, validation, and test partitions provided with the dataset: 30,765 training images, 8,814 validation images, and 4,423 test images. This split design supports model fitting on the training set, hyperparameter and early performance monitoring on validation, and independent reporting on held-out data. Using a stable split across baseline and ablation runs ensured fair comparison between configurations. Results and model behavior were interpreted with attention to split-level class distribution, since imbalance can strongly influence recall for rare PPE categories.

3. TRAINING DETAILS

Training was conducted using a YOLOv8n detector as the primary deep learning baseline, selected for practical speed-accuracy trade-offs in CPU-constrained environments. Experiments were organized as baseline and ablation runs, with controlled changes to data fraction and image size. The training process emphasized reproducibility through fixed seeds, documented run names, and script-based execution. Because deployment target conditions may be resource-limited, a compact reproducible profile was included for faster end-to-end verification.

3.1 Hyperparameters

Baseline settings used a lightweight profile centered on YOLOv8n, with values such as epochs in the low range, moderate batch size, and fixed input resolution for stable iteration speed. Typical runs used confidence and IoU defaults from the detector framework during inference, while training relied on framework-managed optimizer and augmentation defaults. Ablations specifically changed fraction and image size to test whether additional data exposure or higher-resolution features improved detection quality. Across trials, these settings revealed precision-recall trade-offs rather than uniformly improved performance.

3.2 Training Strategy

The strategy followed an iterative experimental cycle: establish a baseline, run controlled ablations, compare metrics, and analyze errors by class and scenario. Instead of maximizing a single metric, the team evaluated practical usefulness for safety monitoring, where missed detections are especially costly. This motivated closer attention to recall and non-compliance classes, not only aggregate precision. The model is therefore positioned as a decision-support detector with human review, rather than a fully autonomous compliance enforcer.

3.3 Compute Resources

Experiments were run in a Python environment on Windows with CPU-based training and inference. No dedicated CUDA GPU was required for the documented baseline pipeline, enabling reproducibility on commodity hardware. This setup supports accessibility for course or prototype settings, though it limits training throughput and may constrain larger model variants. Real-time webcam demonstration was still feasible with the lightweight model and tuned runtime settings.

3.4 Reproducibility Steps

Reproducibility is supported through pinned dependencies, environment configuration files, and script-based commands for data checks, training, and evaluation. A single-command run profile is available to execute a compact training pipeline with deterministic settings and fixed run metadata. The workflow includes dataset sanity checks, run naming conventions, and generated artifacts for direct comparison across experiments. Documentation also links model outputs, ablation tables, and ethics/model-card materials for end-to-end traceability.

4. MODELING

4.1 Model Architecture

The core architecture of the proposed PPE detection system is based on **YOLOv8n**, a one-stage object detection model optimized for efficient real-time object localization and classification [7]. As a one-stage detector, YOLOv8n performs detection and classification in a single forward pass, eliminating the need for a separate region proposal stage and thereby significantly reducing computational overhead and inference time [8]. By utilizing this streamlined approach, the system can bypass the latency issues inherent in multi-stage architectures, allowing it to identify compliance and non-compliance classes in a single computational step suitable for real-time industrial monitoring [9].

The model processes input images through a convolutional backbone that extracts hierarchical spatial features, followed by a feature aggregation mechanism to enhance multi-scale representation. This is particularly important for PPE detection, where objects such as helmets, vests, and gloves may appear at varying sizes and positions. The detection head predicts bounding box coordinates, objectness scores, and class probabilities for each instance, enabling end-to-end learning of both localization and classification tasks.

YOLOv8n's lightweight architecture allows it to operate efficiently in constrained computational environments while maintaining competitive accuracy, making it suitable for real-time safety monitoring with low latency. The output space includes both PPE presence classes (e.g., helmet, safety vest) and non-compliance classes (e.g., missing helmet), enabling the system to generate actionable safety signals directly and simplifying downstream decision-making without additional rule-based logic.

4.2 Component Breakdown

The full system is multi-component by design: CNN for visual detection, NLP for human-readable incident communication, and RL for alert-policy experimentation. These modules are loosely coupled so each can be evaluated independently while still supporting end-to-end workflow integration. The architecture

supports image and stream inputs and produces both structured detections and textual summaries.

4.2.1 CNN Components

The CNN component handles object detection, bounding box generation, class prediction, and confidence scoring. It is trained on PPE and non-PPE labels and evaluated using detection metrics such as mAP, precision, and recall. Inference supports still images, directories, video, and live webcam streams. The detector outputs also serve as input signals for downstream summary and alert modules.

4.2.2 NLP Components

The NLP component converts raw class-count detections into concise natural language summaries for operators. It highlights observed PPE usage patterns and generates short alert text for potential non-compliance conditions. This improves usability by reducing the cognitive load of interpreting raw detection logs in real time. The current implementation is template-driven and can be extended with richer context-aware language generation.

4.2.3 RL Components

The RL component is implemented as an experimental alert-policy prototype to explore reward-driven decision behavior. It models simplified action choices related to when and how to issue alerts based on observed detection states. The objective is not full autonomous control, but policy experimentation that can inform future triage logic. Logged reward history and learning curves provide evidence for analysis rather than production deployment claims.

5. EVALUATION

Evaluation combined quantitative metric reporting and qualitative error interpretation across baseline and ablation runs. The central goal was to assess whether the system is reliable enough for assisted monitoring workflows. Results were interpreted with emphasis on safety relevance, especially the impact of false negatives. This framing aligns evaluation with operational risk rather than benchmark score alone.

5.1 Evaluation Metrics

Primary metrics included mAP at IoU 0.5, mAP at IoU 0.5 to 0.95, precision, and recall. mAP captures detection quality across thresholds, while precision and recall expose trade-offs between false alarms and missed detections. For this project context, recall is especially important because missed non-compliance events can reduce safety impact. Runtime behavior and practical latency were also considered for webcam feasibility.

5.2 Baseline

The baseline YOLOv8n run produced moderate precision but low recall, indicating the model can identify some objects confidently yet still misses a substantial portion of true instances [7]. This establishes a functional starting point for assisted detection but not high-confidence autonomous enforcement, a limitation frequently observed in lightweight models optimized for speed over exhaustive sensitivity in dense industrial scenes [8]. Baseline metrics also provided a critical reference for all subsequent ablation conclusions, aligning with recent experimental standards for evaluating deep learning model iterations [9]. In parallel, a non-deep-learning baseline was included to satisfy methodological comparison requirements and confirm the performance gains provided by learned visual features compared to traditional computer vision approaches [10].

5.3 Ablation Studies

Ablation experiments were conducted to assess the impact of two key factors on model performance: increasing the fraction of training data and raising input image resolution. These ablations were designed to evaluate how changes in training exposure and input quality influence the detection accuracy and reliability of the PPE monitoring system.

In the first ablation, the training data fraction was increased by including a **larger subset of labeled images** during model training. This modification consistently led to improvements in mAP@0.5 across multiple runs, demonstrating that broader exposure to diverse examples enhances the model's ability to generalize to unseen worksite scenarios. This result highlights the importance of dataset size in achieving robust detection performance, particularly for safety-critical applications where missed detections could have serious consequences.

The second ablation focused on increasing the **input image resolution**, aiming to provide the model with more detailed visual information. While this adjustment resulted in a notable increase in precision during one experimental run, it did not yield a proportional improvement in recall. This outcome reflects a classic precision-recall imbalance: the model became more confident in its positive predictions, reducing false positives, but still missed a comparable number of actual PPE instances. This finding indicates that higher-resolution inputs alone are insufficient to fully address missed detections, emphasizing the need for strategies that prioritize recall in future system iterations.

5.4 Error and Slice Analysis

To better understand the limitations of the PPE detection system, an error and slice analysis was conducted to identify patterns in model mispredictions. The analysis indicates that missed detections are concentrated in specific challenging slices, including:

- **Occluded objects**, where PPE was partially blocked by other workers or equipment
- **Small targets**, such as gloves or partially visible helmets
- **Cluttered scenes**, where multiple overlapping objects or complex backgrounds made feature extraction difficult
- **Underrepresented classes**, reflecting an imbalance in the dataset where some PPE types had fewer examples

This pattern aligns with the previously observed recall bottleneck, indicating that the model tends to miss relevant instances under difficult conditions, and underscores the risk of class imbalance affecting overall system reliability.

To address these challenges, several practical mitigation strategies are proposed:

1. **Targeted data enrichment**, by collecting or generating additional examples for slices where performance is poor
2. **Class-aware sampling**, to balance the representation of underrepresented PPE classes during training
3. **Threshold tuning**, specifically optimizing detection confidence thresholds for safety-critical classes to prioritize recall over precision where necessary

Furthermore, slice-based reporting is recommended as a standard practice for each release cycle. This ensures that performance evaluation goes beyond aggregate metrics such as overall mAP and highlights weaknesses that could be masked in global

statistics. Maintaining slice-level monitoring will allow the system to be continuously refined, particularly in safety-critical deployment scenarios where missed detections could have serious consequences.

5.5 Calibration

The current system uses detector confidence scores directly and has not yet completed a full post-hoc calibration study. As a result, confidence values should be interpreted as ranking signals rather than perfectly calibrated probabilities. A recommended next step is class-conditional threshold calibration using validation data to optimize recall for high-risk non-compliance classes while controlling false positives. Reliability plots and threshold sweep reports should be added to future evaluations for operational tuning.

6. ETHICS AND POLICY

The project is designed as a safety-assistive tool with explicit constraints against identity-focused surveillance use. Ethical framing emphasizes harm reduction, human oversight, and responsible deployment boundaries. Policy choices prioritize transparency, conservative automation claims, and documented limitations.

6.1 Risks

Key risks include false negatives in safety-critical contexts, false positives that may cause unnecessary intervention, and misuse for worker surveillance beyond PPE monitoring. Additional risks include domain shift when camera conditions or site environments differ from training data. There is also fairness risk if certain contexts or worker groups are underrepresented in source data. These risks can affect trust, safety outcomes, and organizational adoption quality.

6.2 Mitigation Strategies

Mitigations include human-in-the-loop review before enforcement actions, confidence-aware triage, and periodic model re-evaluation on updated site data. Operationally, the system should log alerts and outcomes for audit and continuous improvement. Thresholds can be tuned to prioritize recall on critical violations while monitoring false alert burden. Governance should include clear usage policy, incident response procedures, and scheduled bias/performance checks.

6.3 Intended Use and Limitations

Intended use is assisted PPE compliance monitoring in image and video workflows, including live demo environments. The system is not intended for identity recognition, disciplinary automation, or standalone enforcement decisions. Current performance supports triage and situational awareness but remains limited by recall, particularly in difficult visual conditions. Human supervision is required for final judgment and action.

6.4 Privacy Considerations

The pipeline focuses on PPE object detection rather than personal identity attributes. Even so, deployment in real workplaces may involve sensitive visual data, requiring retention limits, access control, and secure handling practices. Data minimization principles should be applied, and only outputs needed for safety operations should be stored. Users and workers should be informed of system purpose and boundaries through clear policy communication.

6.5 Fairness and Bias Checks

Fairness checks should evaluate class-level and slice-level performance across diverse environments, lighting conditions, and camera setups. Bias signals may appear as uneven recall across specific contexts or underrepresented class combinations. Monitoring should include periodic re-benchmarking and targeted retraining when disparities are observed. Documentation of these checks should be included in release notes for accountability.

7. MODEL CARD

The model card summarizes architecture, scope, performance, constraints, and safe-use guidance for stakeholders. It functions as a transparency artifact linking technical behavior to operational expectations. This supports responsible handoff from development to deployment teams.

7.1 Model Overview

The PPE detection system is built on YOLOv8n, a one-stage object detector trained for multi-class PPE recognition and non-compliance identification in dynamic workplace imagery. The model processes both static images and live video streams, producing labeled bounding boxes with associated confidence scores for real-time localization and classification of safety equipment. In addition to core visual detection, the system incorporates a lightweight NLP-based summary module to translate detection outputs into concise textual compliance reports, enhancing interpretability for operators and supervisors. A reinforcement learning (RL) policy prototype is also integrated to explore adaptive alert strategies, providing an experimental framework for prioritizing safety-critical detections.

The architecture is optimized for practical inference speed in constrained environments. YOLOv8n's lightweight backbone and single-stage design enable low-latency processing without compromising accuracy, making the system suitable for deployment on edge devices or real-time monitoring stations. By combining visual detection, textual summarization, and adaptive alerting, the system functions as a comprehensive assistive tool for workplace safety monitoring, effectively balancing performance, interpretability, and deployment feasibility.

7.2 Performance Summary

The performance of the PPE detection system demonstrates its practical feasibility for assisted safety monitoring in dynamic worksite environments. Quantitative evaluation indicates that the model can reliably identify personal protective equipment, supporting workflow augmentation and real-time alerts. However, recall remains the primary limitation, as the model occasionally misses PPE instances in challenging conditions such as occlusion, small objects, or cluttered scenes.

Ablation studies further clarify the factors influencing performance. Increasing the training data fraction consistently improved mAP@0.5, showing that broader exposure enhances generalization. Increasing input image resolution improved precision in some runs but did not proportionally increase recall, reflecting a precision–recall trade-off where confidence in positive predictions rises without fully addressing missed detections.

Overall, these findings suggest that the system is effective for supporting human monitoring and compliance workflows but is not yet robust enough for fully autonomous safety enforcement. Production-grade reliability will require continuous retraining with enriched datasets, class-aware sampling, and threshold optimization. This performance assessment underscores that the model is best deployed as an assistive tool, while providing a path

for future improvements in real-world workplace safety applications.

7.3 Deployment Guidelines

Deploy as a decision-support layer in workflows where trained safety personnel validate alerts. Configure confidence thresholds per class and context, with conservative defaults for high-risk non-compliance cases. Track false negatives and false positives in operation and feed findings back into retraining cycles. Use staged rollout with pilot-site evaluation before broader adoption.

7.4 Disclaimers

The system does not guarantee detection of every safety violation and should not be used as sole evidence for punitive action. Performance may degrade outside the data distribution seen during training. Reported metrics are environment-dependent and must be revalidated for each deployment context. Human oversight remains mandatory.

8. DEPLOYMENT AND USAGE

Deployment is designed for practical reproducibility using documented scripts and environment files. The workflow supports both offline evaluation and live webcam demonstration. Operational use should follow safety, privacy, and governance controls from the ethics section.

8.1 Environment Setup

Set up a clean Python environment and install dependencies from the provided requirements specification. Verify that detection libraries and model weights are available before runtime. Confirm dataset paths and annotation structure if retraining or evaluation is required. For stable replication, use the documented scripts and fixed run naming conventions.

8.2 How to Run

The system can be executed from the project root using script-based commands. Dependencies are first installed with **python -m pip install -r requirements.txt**, after which the dataset may be prepared using **python download_data.py --source placeholder --output data** for a local placeholder setup or **python download_data.py --source roboflow --output data** for Roboflow-based retrieval.

Model training is then performed with **python train.py --epochs 10 --batch 8 --img 512 --data data/data.yaml**, and experiment summaries can be generated using **python scripts/build_ablation_table.py**. For inference, the detector supports multiple input modalities: a single image via **python detect.py --model team_webcam_best.pt --source path/to/image.jpg,** a directory via **python detect.py --model team_webcam_best.pt --source data/images/test/**, a video file via **python detect.py --model team_webcam_best.pt --source path/to/video.mp4**, and real-time webcam input via **python.exe detect.py --source 0 --model best.pt --no-webcam-fast --webcam-terminal-log --webcam-log-every 1 --img 640 --vid-stride 2 --max-det 100 --conf 0.25 --source 0**.

A compact reproducibility profile is also available through bash run.sh or make repro. Prior to inference, the selected checkpoint should be verified to ensure class compatibility; for webcam operation, camera permissions must be enabled and other camera-consuming applications must be closed. Execution can be terminated by pressing q in the display window or **Ctrl+C** in the terminal.

8.3 Demo Instructions

Start the detector in webcam mode using the trained checkpoint and monitor live overlays for PPE and non-compliance labels. Keep a controlled test scenario with visible hardhat, vest, and mask examples to demonstrate class behavior clearly. Use the generated NLP summaries and alerts to explain how detections are translated into operator-facing messages. End the session with a short discussion of limitations, especially recall-related misses, and emphasize human-in-the-loop use.

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